Hyperspectral Image Super-Resolution Network Based on Cross-Scale Nonlocal Attention

Shuangliang Li, Yugang Tian, Cheng Wang, Hongxian Wu, and Shaolan Zheng

***Abstract*— Hyperspectral image (HSI) super-resolution generally means the fusion of low-spatial-resolution HSI (LRHSI) and high-spatial-resolution multispectral/panchromatic image (HRMPI) to get high-spatial-resolution HSI (HRHSI). Existing fusion methods have not sufficiently considered the huge spectral and spatial resolution difference between the LRHSI and HRMPI. In addition, most deep learning (DL)-based methods that adopt the convolutional neural network (CNN) structure are limited by its local feature learning, and it is difficult to exploit the global dependence of image features. To fully adapt to the huge modality difference between LRHSI and HRMPI and release the limitation of local feature learning, we design the cross spectral-scale and shift-window-based cross spatial-scale nonlocal attention network (CSSNet) to effectively fuse the LRHSI and HRMPI. These two networks could explicitly learn the spectral and spatial correlations between two input images. These correlations are then used to reconstruct the HRHSI feature, which makes the obtained HRHSI feature to maintain the spectral and spatial feature consistency with the input images. Finally, a "feature aggregation module" is designed to aggregate the image features from these two networks and output the fused HRHSI. Extensive experimental results on both HM-fusion [fusion with multispectral (MSI)] and HP-fusion (fusion with panchromatic (PAN) image) tasks demonstrate CSSNet's state-of-the-art (SOTA) performance compared to other fusion methods. The codes could be available at https://github.com/rs-lsl/CSSNet.**

***Index Terms*— Cross-scale, hyperspectral image (HSI) fusion, huge resolution difference, nonlocal attention.**

I. Introduction

THE hyperspectral image (HSI) with hundreds of spectral bands provides rich spectral information for different materials and objects. Also, this has made it utilized in many real-world applications, such as classification [1], [2], spectral unmixing [3], [4], and mineral exploitation [5].

However, due to the limitation of satellite imaging sensors, HSI always suffers from the coarse spatial resolution, which leads to the loss of detailed information, including tiny spatial textures and details. To effectively improve the spatial resolution of the low-spatial-resolution HSI (LRHSI), many HSI super-resolution (SR) methods have been proposed to fuse the LRHSI and high-spatial-resolution multispectral/panchromatic image (HRMPI) to obtain the high-spatial-resolution HSI (HRHSI) with both the high spatial and spectral resolutions [6], [7], [8], [9], [10], [11], [12], [13], [14].

The fusion method between LRHSI and HRMPI could be summarized as two main classes, including variational optimization (VO) model-based [16], [17], [18], [19], [20] and deep learning (DL)-based methods [9], [11], [12]. For the first class, they usually formulate the fusion process as a constrained optimization problem under the specific prior (e.g., sparse and low-rank priors) and alternatively optimize the desired variables until convergence. The popular methods in this class include coupled nonnegative matrix factorization (CNMF) [19], Bayesian sparsity promoted Gaussian prior (Bayesian sparse) [21], HySure [18], and Bayesian naive Gaussian prior (Bayesian naive) [22]. Even with the satisfactory fusion performance they could achieve, most of these methods suffer from the high computational cost due to the iterative parameter optimization process [14], [23].

Recently, DL-based fusion methods have been developed explosively in the image SR and fusion fields due to their

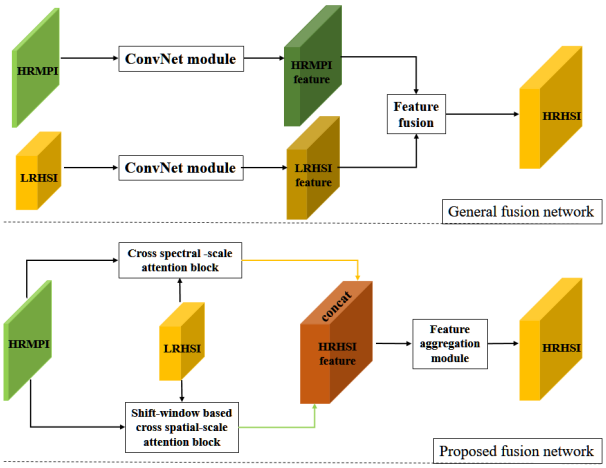


Fig. 1. Comparison between the network structure of the general fusion network and the proposed fusion network. "ConvNet module" represents the prevalent CNN network structure for extracting image features from input images. "concat" means the channelwise concatenation operation on the image features resulting from two cross-scale attention blocks. Then, the DWconv [15] with kernel sizes 11 and 1 is used to aggregate the image features and output the fused HRHSI.

Manuscript received 28 January 2023; revised 16 March 2023 and 6 April 2023; accepted 18 April 2023. Date of publication 26 April 2023; date of current version 8 May 2023. This work was supported by the Key Laboratory of Natural Resources Monitoring in Tropical and Subtropical Area of South China, Ministry of Natural Resources, under Grant 2022NRM003. *(Corresponding author: Yugang Tian.)*

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Digital Object Identifier 10.1109/TGRS.2023.3269074

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powerful feature learning ability [24], [25], [26], [27], [28], [29], [30], [31]. For example, Jiang et al. [24] proposed a generative adversarial network-based edge-enhancement network for the robust satellite image SR. Xiao et al. [27] designed a deformable convolution alignment module, a temporal grouping projection fusion strategy, and a temporal attention module for the video SR task. Actually, many fusion methods use the convolutional neural network (CNN) to encode the input images and fuse them in the feature space. Then, they output the fused HRHSI by decoding operation, as shown in the upper part of Fig. 1. Xiao et al. [28] designed the detail extraction network and the spatiospectral fusion network to obtain the fused HRHSI. Zheng et al. [29] proposed an edge-conditioned feature transform network to fuse the LRHSI and HRMPI, which is composed of feature extraction, feature fusion, and the last reconstruction module.

However, these fusion methods are mostly borrowed from the multispectral (MSI) image pansharpening field without considering the large modality difference between the LRHSI and HRMPI. For example, the ZY-1 02D satellite obtains the LRHSI of pixel size 30 m with 166 bands and the MSI of size 10 m with eight bands and the panchromatic (PAN) image of size 2.5 m [32]. This extremely large spectral and spatial resolution difference between LRHSI and HRMPI increases the ill-posedness of the fusion problem greatly. A few studies have focused on this large resolution difference. For example, Li et al. [33] designed an interpretable detail injection block to inject the spatial details by justifying the data distribution to the correct "fish" distribution. However, the "fish" distribution may not be completely correct for different satellite images and lacks the theoretical proof. Therefore, it is essential to design an HSI fusion network that can adapt to the huge modality difference between LRHSI and HRMPI and alleviate the ill-posedness of the fusion task effectively.

Furthermore, most of the fusion networks are constructed based on the prevalent CNN structure. However, its limited local feature learning ability hinders the exploration and utilization of the long-range dependence of the image features. Recently, the transformer structure [34] has been developed in the computer vision fields such as SR [35], [36]. Its main idea is to use the nonlocal self-attention to enhance the image features. For example, Lu et al. [36] proposed an SR transformer for single-image SR, which consists of a CNN backbone and a lightweight transformer network. Yu et al. [37] devised a nonlocal attention mechanism that enables video SR without frame alignment and a memory-augmented attention module to memorize the video details. Yi et al. [38] proposed a progressive fusion network and an improved nonlocal operation for the video SR.

In the field of image fusion, several nonlocal attention-based methods have also been proposed. For example, Bandara and Patel [39] adopted the multihead attention module in the transformer to enhance the LRHSI feature using the attention map computed from the HRMPI. Also, these methods require matching the feature size of the input images before computing the pixelwise nonlocal attention (similarity matrix). However, this matching operation (e.g., bicubic upsampling method) may introduce additional noise or errors that degrade the resulting image quality. In addition, their nonlocal attention is computed on the pixel level, which may lose the local feature information of the image.

To effectively address the above problems, we make three main contributions. First, considering the large difference in spectral resolution between the LRHSI and HRMPI, we design a cross spectral-scale nonlocal attention (CSNA) module. This module first computes the spectral correlation between the HSI and MPI on the low-resolution scale. It then maps this correlation to the HRMPI feature to reconstruct the HRHSI feature. This module could sufficiently capture the spectral correlation between two input images on the original spectral dimension and improve the spectral fidelity of the reconstructed HRHSI.

Second, to adapt to the spatial resolution gap between the LRHSI and the HRMPI, we propose a shift-window-based cross spatial-scale nonlocal attention (SW-CSNA) module. It could map the cross spatial-scale global attention extracted from high-level features to the original LRHSI feature and reconstruct the HRHSI feature. In addition, to promote the correctness of the encoded high-level features, we proposed a fast intraclass difference-aware (ida) loss function. This could benefit the designed encoding block to extract the semantic information (high-level feature) of the input image as correctly as possible.

Third, we test the proposed fusion framework on two kinds of fusion tasks, including fusion between LRHSI and HRMSI (HM-fusion) and between LRHSI and PAN images (HP-fusion). Also, the experimental results on simulated and real datasets all show that our method achieves the state-of-the-art (SOTA) performance, demonstrating the superiority and robustness of the proposed fusion network.

The remaining part of this article is organized as follows. Section II introduces the related work with the research issues, including HSI SR that fuses with the MSI and PAN image and the nonlocal attention. Section III describes the designed network modules and loss functions. Results of the ablation study and comparative experiments are given in Section IV. Finally, the conclusion is given in Section V.

II. Related Work

A. Hyperspectral and MSI Image Fusion Methods

The fusion-based HSI SR methods between HSI and MSI mainly include VO- and DL-based. VO-based methods generally formulate the fusion process as an optimization problem and impose handcrafted priors to constrain the solution. For example, Akhtar et al. [6] extracted the spectral basis from the LRHSI and imposed the sparsity and nonnegativity constraints on the spatial component. Li et al. [7] regarded the image as a 3-D tensor and incorporated a sparse regularizer to exploit the high spatial–spectral correlations in the HRHSI. Yokoya et al. [19] proposed a CNMF fusion method, in which LRHSI and HRMSI are alternately decomposed into abundance and endmember components based on the spectral linear mixture model.

In order to preserve spatial and spectral smoothness, some tensor factorization-based methods have been proposed to solve the fusion problem. For example, Zhang et al. [40]

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designed a low-rank Tucker decomposition model that combines two graphs to fuse the LRHSI and HRMSI. Kanatsoulis et al. [41] proposed a coupled tensor factorization method that improves the identifiability of the SR image under the realistic and mild conditions.

DL-based HSI SR methods have been developed explosively in recent years. Some typical network structures are adopted to solve the fusion problem, such as Resnet [42], Dense net [43], and U-net [44]. For example, Han et al. [10] designed a multiscale fusion network that gradually modified the image feature sizes. Zhang et al. [11] combined the residual channel attention and Dense-net to learn the spatial–spectral correlation and achieved great fusion performance. Consider the observation models of the HRMSI and LRHSI, Xie et al. [9] constructed an interpretable image fusion network, which unfolded the constructed iterative algorithm using different network layers. In addition, FusionNet [45] was proposed to fuse the spatial and spectral information of the LRHSI and HRMSI by using a variational probabilistic autoencoder.

B. Hyperspectral and PAN Image Fusion Methods

Due to the extremely ill-posedness of the fusion problem between the HSI and the PAN image, the traditional methods, including component substitute (CS) and multi-resolution analysis (MRA)-based pansharpening methods, are hard to achieve great performance. Also, the dominant methods are mainly VO- and DL-based, especially the latter.

For the VO-based fusion method, they usually formulate the fusion process as the variable optimization problem under the constraint of the specific priors. For example, Yokoya et al. [19] decomposed the input images into the spectral basis and coefficients to get the fused result. Wei et al. [21] formulated the fusion problem within a Bayesian estimation framework and introduced an appropriate prior distribution that exploits the geometric considerations. Simões et al. [18] formulated the HSI fusion problem as the minimization of a convex function that contains two quadratic data-fitting terms and an edge-preserving regularizer. Despite the satisfactory fusion performance they have achieved, most of these methods suffer from the high computational cost.

DL-based methods have been developed explosively in recent years due to their powerful feature learning ability. Many hyperspectral and PAN image fusion networks have been designed and achieved great performance. For example, Zheng et al. [46] exploited the deep image prior (DIP) to upsample the LRHSI and proposed the spatial and spectral attention network to inject the PAN details into the LRHSI. Then, Bandara et al. [47] used the improved DIP and residual structure to obtain the reconstructed HRHSI with a learnable spectral response function (SRF). Qu et al. [48] designed a dual-branch detail extraction network that could sharpen the LRHSI with any number of spectral bands by the pretrained model. However, these methods are based on the CNN with limited local feature learning and lack the ability to explore the long-range dependence of the image feature.

Recently, based on the transformer module [49], Bandara and Patel [39] proposed a HyperTransformer fusion network, including two separate feature extractors, a multihead attention module, and a spectral–spatial feature fusion module to reconstruct the HRHSI. However, the nonlocal attention in this study is computed on the pixel level and may ignore the local texture and context information of the image feature.

C. Patch Recurrence Property and Nonlocal-Based Attention

Image patches tend to recur within and across scales of the same image, which is called the patch recurrence property [50], [51]. This property has been widely used in the image SR field to enhance the HR patch using similar patches in different resolutions' images. In the pioneering study, Glasner et al. [50] integrated the methods of multiple images SR and example-based SR to exploit repeating patches within and across multiscales of the image. Furthermore, Freedman and Fattal [52] effectively extracted the patches from the localized regions of the input image, which could reduce the computational complexity.

This patch recurrence property within and across scales represents the long-range dependence of the image patches, which is similar to the transformer's nonlocal self-attention [34]. Also, some studies have used the nonlocal attention module to super-resolve the LR image. For example, Liu et al. [53] proposed a nonlocal recurrent network to incorporate the nonlocal operations into a recurrent neural network for image restoration with fewer parameters. Dai et al. [54] proposed a channel attention module to adaptively rescale the channel-wise features and a nonlocally enhanced residual group to capture long-distance spatial information. Recently, Bandara and Patel [39] proposed a HyperTransformer fusion network, including two separate feature extractors, a multihead attention module, and a spectral–spatial feature fusion module to reconstruct the HRHSI.

III. Methodology

A. Overall Fusion Framework

The proposed fusion framework is shown in the bottom of Fig. 1. To adapt to the large spectral and spatial resolution difference between two input images, we design the cross spectral-scale and cross spatial-scale blocks to sufficiently learn the nonlocal spectral and spatial relations of the input images on low-dimensional features. These relations are then mapped into the high-dimensional feature to reconstruct the HRHSI feature. Then, a "feature aggregation module" follows to integrate the HRHSI features from these two cross scale blocks and output the fused HRHSI.

Unlike most fusion networks that extract the latent feature from the input images and fuse them, we explicitly learn the spectral and spatial correlations between two input images. These correlations are then used to reconstruct the HRHSI, which makes the fused HRHSI maintain the spectral and spatial feature consistency with the input images and improves the spectral and spatial fidelity of the fused HRHSI.

B. CSNA Block

In fact, in some spectral mapping function (from HSI to MPI) estimation methods, they assume the bandwise linear

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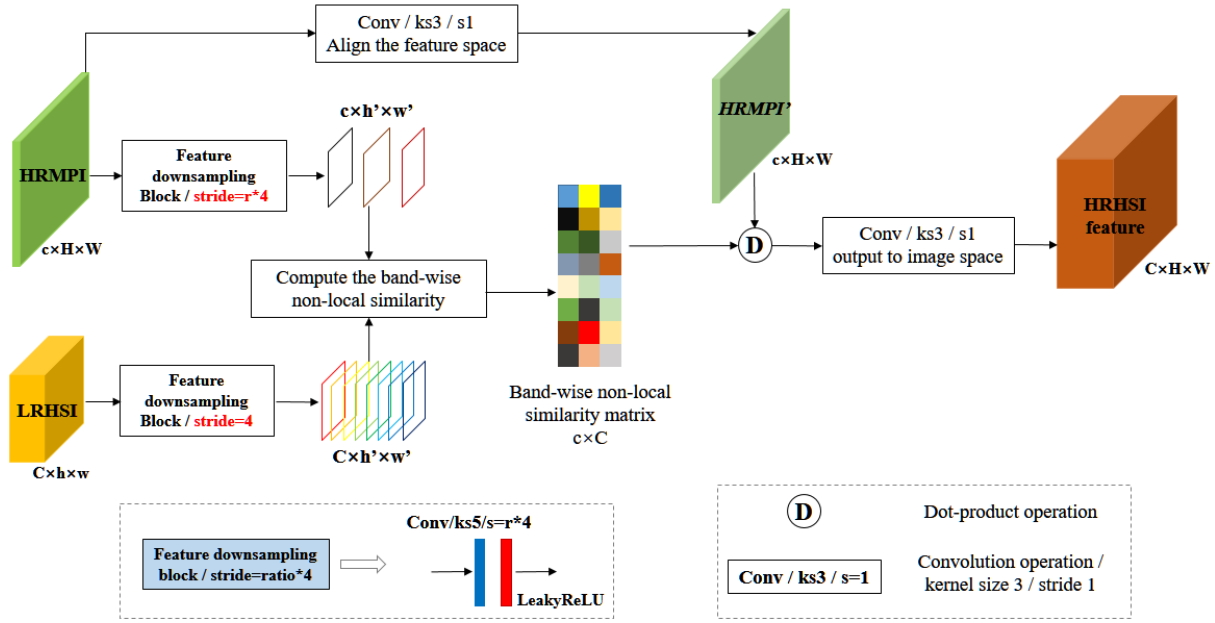


Fig. 2. Network structure of the proposed CSNA block. The "feature downsampling block" includes a stride convolution and a LeakyRelu activation function. "r" means the spatial resolution ratio between HRMPI and LRHSI.

relation between HSI and MPI at the same spatial resolution. Also, they use the downsampled images to estimate the mapping function and apply it to the high-resolution scale [55], which achieves great performance. Therefore, the spectral band correlation (between HSI and MPI) on the LR scale is approximately the same as that on the HR scale

$$\psi\left(M^i_{h\times w}, H^j_{h\times w}\right) = \psi\left(\mathbb{M}^i_{H\times W}, \mathbb{H}^j_{H\times W}\right) \tag{1}$$

where $M^i_{h\times w}$ represents the **i**th band of the low resolution MPI and $\mathbb{H}^j_{H\times W}$ represents the **j**th band of the high-resolution HSI. Their spatial sizes are $h \times w$ and $H \times W$ in LR and HR scales, respectively. $\psi(\cdot,\cdot)$ means the function to compute the correlation as

$$\psi\left(\mathbb{M}^i_{H\times W}, \mathbb{H}^j_{H\times W}\right) = \theta\left(\mathbb{M}^i_{H\times W}\right) \cdot \delta\left(\mathbb{H}^j_{H\times W}\right) \tag{2}$$

where $\theta(\cdot)$ and $\delta(\cdot)$ are feature transform functions without changing the number of spectral bands and $\cdot$ means the dot-product operation to compute the correlation coefficient between $\mathbb{M}^i_{H\times W}$ and $\mathbb{H}^j_{H\times W}$.

To map the bandwise global similarity from the LR scale to the HR scale, we design the CSNA, as shown in Fig. 2. In this block, we first downsample both input images by the single stride convolution from $H \times W$ and $h \times w$ to the same size $\rightarrow$ $h' \times w'$ (we set h' and w' equal to $h/4$ and $w/4$, respectively). This downsampling operation could get the discriminative features and reduce the complexity of the following similarity computation. Note that we maintain the original spectral band number through this block, which could greatly preserve the spectral self-correlation and cross correlation of the input images.

Then, we compute the bandwise similarity matrix

$$\phi(M_{c\times h'\times w'}, H_{C\times h'\times w'}) = (M_{c\times h'w'})(H_{C\times h'w'})^T \Rightarrow S_{c\times C} \tag{3}$$

where "$S_{c\times C}$" is the resulting similarity matrix, "c" and "C" are the number of channels of HRMPI and LRHSI, respectively, and h' and w' represent the spatial size of the downsampled image features, as shown in the middle part of Fig. 2. Also, ^{T} means the transpose operation. Note that $\psi(.,.)$ in (2) is different from $\phi(.,.)$ in (3). The former is computed on the single-band image, while the latter is computed on the all-band image.

To map the similarity matrix computed on the LR scale to the HR scale, we use the dot-product operation in the spectral dimension to obtain the reconstructed HRHSI feature. Note that we first align the feature space of HRMPI by the convolution operation to get HRMPI. Then, we take the dot-product operation between the similarity matrix and the HRMPI feature

$$f_{\text{softmax}}(S_{c\times C})^T \cdot \mathbb{M}'_{c\times H\times W} \Rightarrow \mathbb{H}'_{C\times H\times W} \tag{4}$$

where $\mathbb{M}'_{c\times H\times W}$ means the HRMPI feature, which has been aligned the feature space, $\mathbb{H}'_{C\times H\times W}$ is the reconstructed HRHSI feature, and $f_{\text{softmax}}(.)$ means the softmax function to normalize the similarity matrix.

C. SW-CSNA Block

To adapt to the spatial resolution gap between the LRHSI and HRMPI and to benefit from both the local and nonlocal feature learning, we propose the SW-CSNA, as shown in Fig. 3. This module unfolds the image into patches and computes the cross-scale nonlocal attention on the patch level to enhance the resulting image feature. Unfortunately, a single patch crop mode may not accurately capture a complete object in a patch. For example, as shown in Fig. 4, if we adopt only one kind of patch crop mode as in Fig. 4(a), an object such as a house may be separated into two patches and lose its semantic

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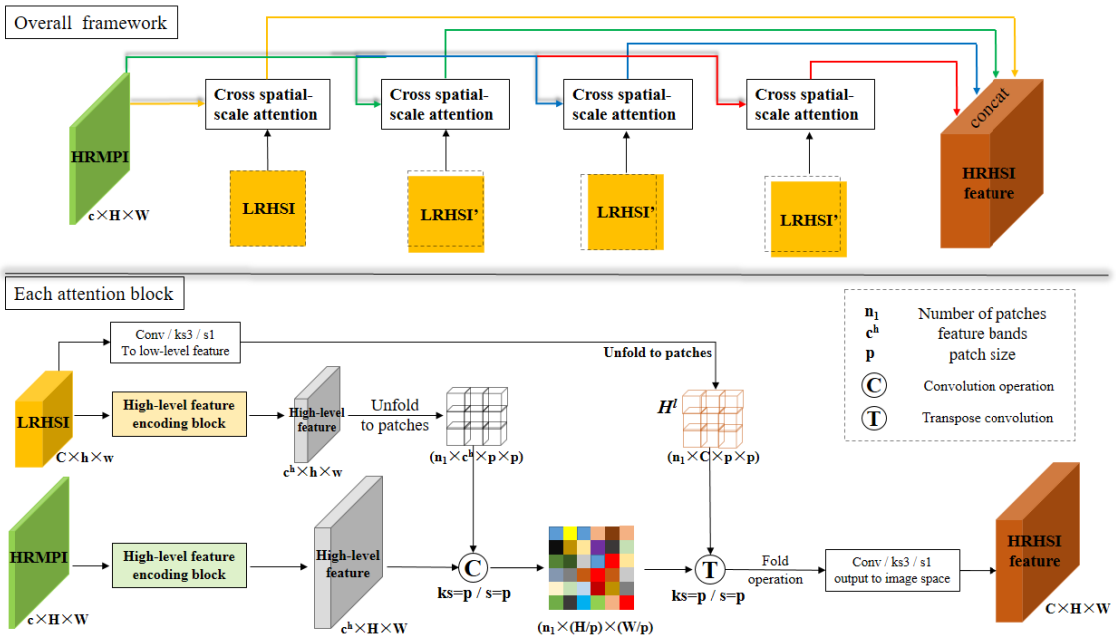


Fig. 3. Network structure of the proposed SW-CSNA block. Four "dashed squares" in the upper part represent the shifted image patches with different shift values. The detailed structure of the "high-level feature encoding block" is shown in Fig. 5. Note that "p" means the unfolding patch size and unfolded patches act as the convolution/transpose convolution kernels.

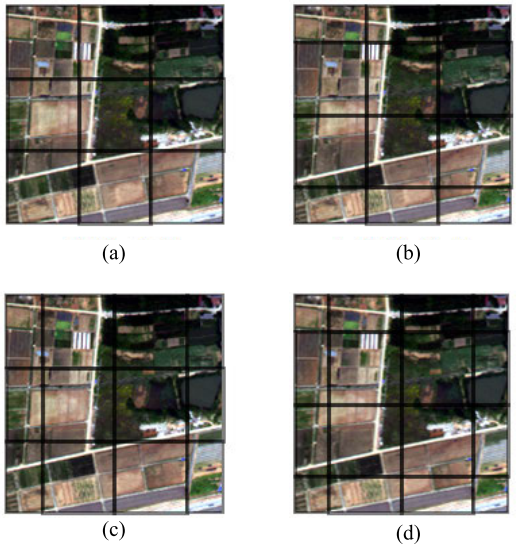


Fig. 4. Demonstration of the "shift window," which means that we shift the patch crop window with the different shift values. (a) Shift = (0, 0). (b) Shift = (1, 0). (c) Shift = (0, 1). (d) Shift = (1, 1).

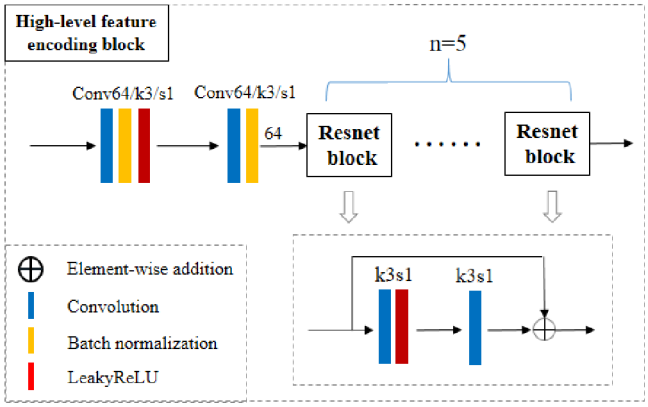


Fig. 5. Detailed network structure of the proposed "high-level feature encoding block".

information. Therefore, to sufficiently adapt to objects with different sizes and positions, as shown in Fig. 4 and the top of Fig. 3, we design a shift-window-based nonlocal attention block that adopts four kinds of patch crop modes. Specifically, we shift the whole input LRHSI in each attention block with different shift values in horizontal and vertical directions. This could accommodate more complete objects into a single patch. Note that the shift value is smaller than the unfolding patch size to capture complete objects as much as possible.

In each attention block, we first measure the unfolded patches' nonlocal similarity between the encoded high-level features of the HRMPI and LRHSI, as shown in the middle of Fig. 3. This similarity is then mapped to the original LRHSI feature to obtain the reconstructed HRHSI feature. Note that the patchwise nonlocal similarity could preserve more local texture information compared to the pixelwise similarity. Also, the combination of local and nonlocal learning is more effective and efficient in improving image feature learning and fusion performance.

Actually, the assumption of the similarity mapping is that the similarity between the image's high-level features is closely related to the similarity between the low-level features. Generally, the high-level feature means the semantic information and the low-level feature represents the shallow spatial texture and spectral feature. For example, the trees in the same category (high-level feature) always have almost the same spatial and spectral features (low-level feature). Therefore, the

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spatialwise nonlocal similarity matrix could be shared between the high- and low-level features

$$\psi\left(\mathbb{M}_{i,j}^{h}, H_{m,n}^{h}\right) = \psi\left(\mathbb{H}_{i,j}^{l}, H_{m,n}^{l}\right) \tag{5}$$

where $\mathbb{M}_{i,j}^{h}$ means the (**i**th, **j**th) unfolded patches of the HRMPI's high-level feature. $H_{m,n}^{h}$ means the (**m**th, **n**th) unfolded patches of the LRHSI's high-level feature. Also, $\mathbb{H}_{i,j}^{l}$ means the (**i**th, **j**th) unfolded patches of the HRHSI's low-level feature. "H" means the LRHSI (these three images cover the same scene). The function $\psi(\cdot,\cdot)$ is the same as (2) to compute the correlation between them.

Therefore, we first obtain the high-level features of LRHSI and HRMPI through two "High-level feature encoding blocks," as shown in the left of Fig. 3. The detailed network structure of the "high-level feature encoding blocks" is shown in Fig. 5. It consists of two cascaded convolution blocks, containing the convolution layer, batch normalization, and LeakyRELU functions. Then, five "Resnet blocks" are used to extract the image's high-level feature. It is expected that the cascade network layers could extract the deep semantic information of the image, which has been verified in many studies [42], [56]. Note that since the input images of each cross spatial-scale attention block only have different shift values, these two "encoding blocks" are shared in these attention blocks. This sharing mechanism not only stabilizes the parameter optimization process but also reduces the model parameters. Then, to compute the similarity matrix, the encoded high-level feature of LRHSI is unfolded into patches

$$H_{c^h\times h\times w}^{h} \rightarrow \text{unfold}() \rightarrow H_{n_1\times c^h\times p\times p}^{h} \tag{6}$$

where $H_{c^h\times h\times w}^{h}$ is the encoded high-level feature of LRHSI, c^h is its band number, and n_1 is the number of patches after the unfold() operation based on the patch size p ($n_1 = (h\times w)/(p\times p)$). Also, the unfolded patches are shown by the "cube grid" in Fig. 3.

To compute the cross-scale patchwise nonlocal similarity with high efficiency, we apply the convolution operation from the unfolded patches in (6) (acting as the convolution kernel) to the high-level feature of HRMPI

$$f_{H_{n_1\times c^h\times p\times p}^{h}}\left(\mathbb{M}_{c^h\times H\times W}^{h}\right) \Rightarrow S_{n_1\times(H/p)\times(W/p)} \tag{7}$$

where $\mathbb{M}_{c^h\times H\times W}^{h}$ is the encoded high-level feature of HRMPI and $S_{n_1\times(H/p)\times(W/p)}$ is the resulting nonlocal similarity matrix. Note that the stride of this convolution operation is set to the patch size p. Then, as shown in Fig. 3, to reconstruct the HRHSI, the low-level feature of the LRHSI (H^l) is unfolded

$$H_{C\times h\times w}^{l} \rightarrow \text{unfold}() \rightarrow H_{n_1\times(C\times p\times p)}^{l}. \tag{8}$$

Finally, we map the similarity matrix $S_{n_1\times(H/p)\times(W/p)}$ to $H_{n_1\times(C\times p\times p)}^{l}$ through transpose convolution operation as inspired by [57] to obtain the reconstructed HRHSI feature

$$f_{H_{n_1\times(C\times p\times p)}^{l}}\left(f_{softmax}\left(S_{n_1\times(H/p)\times(W/p)}\right)\right) \Rightarrow \mathbb{H}_{C\times H\times W}^{l} \tag{9}$$

where $f_{H_{n_1\times(C\times p\times p)}^{l}}(.)$ means to perform the transpose convolution operation using the unfolded kernel $H_{n_1\times(C\times p\times p)}^{l}$ on the normalized $S_{n_1\times(H/p)\times(W/p)}$ with the stride of patch size p.

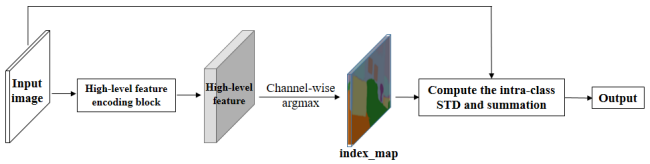


Fig. 6. Illustration of the proposed ida loss function. The computed semantic label is used to index the input image to compute the intraclass STD followed by the summation.

Algorithm 1 Fast ida Loss Function Computation

Require: image (N × c × H × W)
image_h (high level feature of image, N × c^h × H × W)
win_size (the window size to divide the sorted image)
$argmax$(., dim): return the indices of the maximum values along the 'dim'
$argsort$(., dim): return the indices of the sorted values along the 'dim'
std(., dim): compute the 'STD' along the 'dim'
sum(., dim): sum the value along the 'dim'
Ensure: summation of intra-class STD
index_map = $argmax$(image_h, dim = 1)
index_map = index_map.$reshape$(N, HW)
sort_index = $argsort$(index_map, dim = 1)
image = image.$reshape$(N, c, HW)
for i = 0:N **do**
sorted_image[i,:,:] = image[i,:, sort_index[i,:]]
end for
sorted_image = sorted_image.$reshape$(N, c, win_size, (H*W/win_size))
std_image = std(sorted_image, dim = 2)
result = sum(std_image)

D. Feature Aggregation Module

After obtaining the reconstructed HRHSI features from each cross spatial-scale attention block with different shift values, we concatenate them with the HRHSI feature from the cross spectral-scale block. Then, as shown in Fig. 1, the feature aggregation module, including depthwise separable convolution (DWconv) with kernel sizes 11 and 1, is used to integrate the image features and output the fused HRHSI. Note that we add the residual connection [42] from the upsampled LRHSI (through bicubic interpolation) to the final result to accelerate the convergence of the fusion network.

E. Loss Functions

The proposed network is trained using the L_1 distance function

$$L_1 = \left\|\mathbb{H} - \hat{\mathbb{H}}\right\|_1 \tag{10}$$

where $\mathbb{H}$ means the reference image and $\hat{\mathbb{H}}$ is the fused result. This function could effectively alleviate the blurring effect induced by the L_2 loss function. In addition, to promote the structural similarity of the fused HRHSI with the reference image, we add the structure similarity (SSIM) loss function [58] as

$$L_{\text{SSIM}} = 1 - \text{SSIM}\left(\mathbb{H}, \hat{\mathbb{H}}\right). \tag{11}$$

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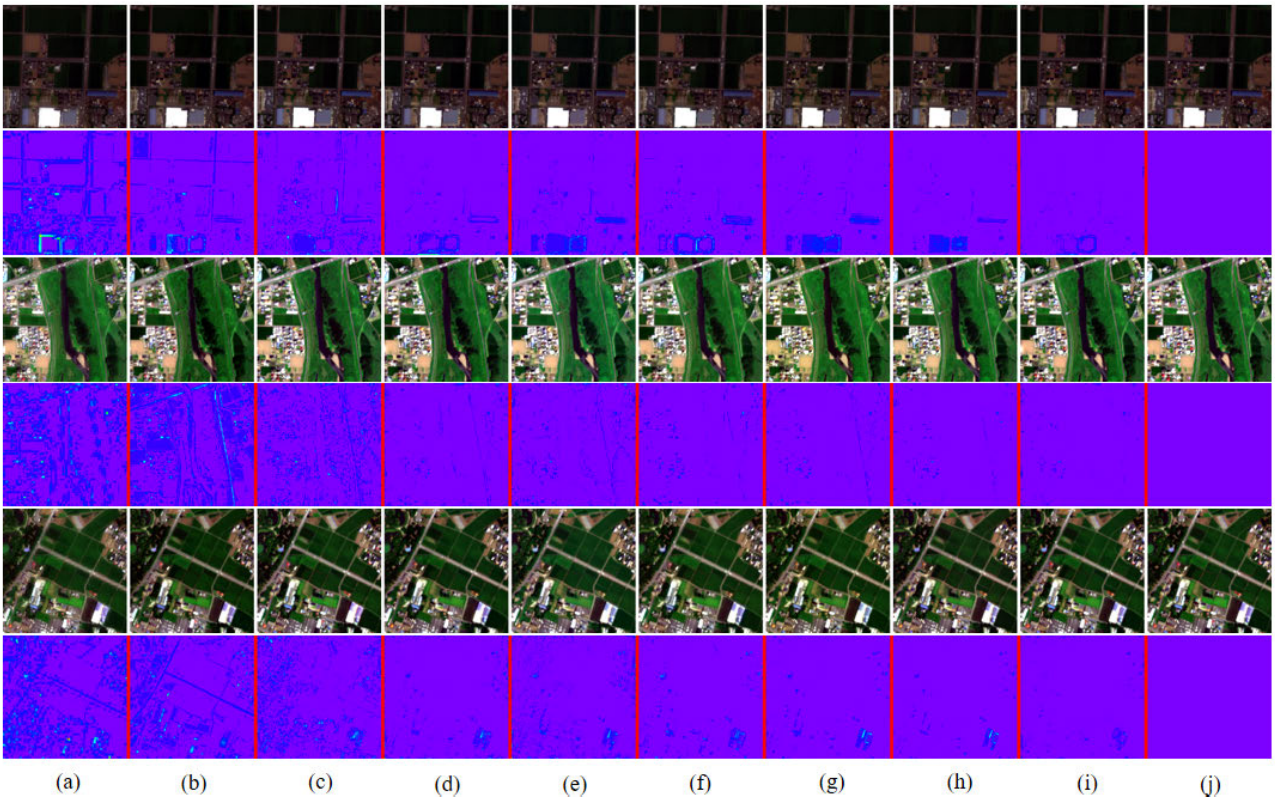


Fig. 7. Fused results (odd row) and error maps (even row) of different methods on Chikusei-m datasets. (a) SFIM. (b) CNMF. (c) SSRNet. (d) HSRNet. (e) EDBIN. (f) Fusformer. (g) MSSJFL. (h) SpfNet. (i) Ours. (j) Ground truth.

Furthermore, to improve the correctness of the high-level features extracted by the "high-level feature encoding block" in the SW-CSNA, we proposed an ida loss function. As shown in Fig. 6, after extracting the high-level feature, the argmax operation along the spectral dimension is used to extract the rough semantic category information as indicated in [59] (we formulate the pseudocode of its fast version in Algorithm 1 with fast computation speed). Then, we compute the intraclass standard deviation (STD) of the original input image in each semantic category. Theoretically, the low-level image features in the same semantic class would be similar enough to result in a small intraclass feature difference. Therefore, it is expected that this loss function could improve the correctness of the extracted high-level features.

This loss function could be simply expressed as

$$L_{\text{ida}} = \sum_{i=1}^{n} f_{\text{STD}}(H_i) \tag{12}$$

where $f_{\text{STD}}(H_i)$ means to compute image's intraclass STD in the **i**th semantic category. Also, we sum all class's STD to get L_{ida} (n is the class number). Note that considering the imbalance of each class and the computational efficiency, the idea of "window" is used to compute the intraclass STD, as in Algorithm 1 (the "window" division is implemented by the "reshape" operation). This means that after sorting the image based on the "index_map," we divide the image by the window size and compute the STD in each window. Note that this loss function is computed on both the original HRMPI and LRHSI images. Also, considering the input image size and categories number, the windows sizes of MPI and HSI are set to 16 and 9 for the HM-fusion task and 32 and 16 for the HP-fusion task, respectively.

Therefore, the overall loss functions are

$$L_{\text{all}} = L_1 + \alpha L_{\text{SSIM}} + \beta L_{\text{ida}} \tag{13}$$

where α and β are the weighting coefficients of the corresponding loss function.

IV. Experiments

A. Datasets

We test the proposed fusion network on two different fusion tasks, including fusion between LRHSI and HRMSI (HM-fusion) and fusion between LRHSI and PAN image (HP-fusion). We select the Chikusei [60], Ziyuan (ZY), and XiongAn [61] datasets for the fusion experiments.

The Chikusei dataset was captured by the Headwall Hyperspectral-VNIR-C imaging sensor over Chikusei, Japan. The original HSI consists of 128 spectral bands. The spatial size is 2517 × 2335 with a resolution of 2.5 m. We generate the simulated LRHSI and HRMSI with five bands (Chikusei-m), LRHSI and PAN image (Chikusei-p) datasets through the Gaussian downsampling, and spectral dimension projection by the SRF of the Worldview2 satellite sensor. Also, the original HSI seems as the reference HRHSI.

The real ZY dataset was acquired by the Ziyuan 1-02D satellite, and it originally includes LRHSI with 166 bands, HRMSI with eight bands, and PAN images. Their spatial

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sizes are 2064 × 2028, 6192 × 6048, and 24 768 × 24 336. Also, due to lack of the HRHSI reference image, we spatially downsample the original LRHSI, HRMSI, and PAN image using the Gaussian kernel according to the Wald protocol [62] to generate ZY-m and ZY-p datasets. Then, the original LRHSI is regarded as HRHSI. Note that due to the low SNR of some bands of the ZY dataset, we select only the first 76 bands of its LRHSI in the fusion experiment.

The XiongAn dataset was captured in Matawan Village of XiongAn New Area, China. This village mainly consists of different kinds of crops. This dataset contains 256 bands with a spatial resolution of 0.5 m. Its height and width are 1580 and 3750, respectively. Using the same operation as the Chikusei dataset, we generate the simulated LRHSI and PAN images.

Note that the spatial downsampling ratios for the HM-fusion and HP-fusion tasks are 3 and 12, respectively. Also, the cropping patch sizes in the training and testing stages are 48 × 48 (LRHSI), 144 × 144 (HRMSI), 16 × 16 (LRHSI), and 192 × 192 (PAN) for these two fusion tasks.

B. Training Details

We use the Adam optimizer to train the proposed network for 500 epochs with the designed loss functions. The weight coefficients α and β in the loss functions are set to 0.1 and 0.01, respectively. The batch size is set to 10 and the learning rate is initialized to $2e^{-4}$, then decreasing by a factor of 0.5 at 200 and 400 epochs. The number of feature bands is all set to 64 unless otherwise specified. The codes are run under the Paddle 2.2.2 framework and the Python 3.7 environment on a single V100 graphic processing unit (GPU) with 32-GB memory.

C. Compared Methods and Performance Metrics

We compare the proposed cross spectral-scale and shift-window-based cross spatial-scale nonlocal attention network (CSSNet) with several SOTA methods, including traditional and DL-based approaches. For the HM-fusion, traditional methods include SFIM [63] and CNMF [19]. Also, DL-based methods include SSRNET [64], HSRnet [12], EDBIN [65], Fusformer [13], MSSJFL [66], and SpfNet [67]. For the HP-fusion, the first class includes GSA [68], SFIM [63], Wavelet [69], and MTF_GLP_HPM [70]. DL-based methods include HSpeSet2 [71], GASN [72], Pgnet [33], and RHDN [73]. For a fair comparison, the training epochs and batch size of these DL-based methods are all adjusted to achieve their best performance.

The quality indices used to measure and compare the performance of different methods include peak signal-to-noise ratio (PSNR) [74], SSIM [58], Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS) [14], spectral angle mapper (SAM) [14], and spatial consistency coefficient (SCC) [14]. Note that the error map is also given to show the visual difference between the fused result and the reference image, as shown in the even rows of the visual results. Also, the error map ranging from 0 to 0.1 is calculated from the average of all bands' absolute difference between the reference and the fused result.

TABLE I

AVERAGE QUANTITATIVE RESULTS ON **ZY-M** DATASET WITH DIFFERENT NETWORK STRUCTURES. "SPECTRAL BLOCK" MEANS THE CROSS SPECTRAL-SCALE BLOCK, WHILE "SPATIAL BLOCK" REPRESENTS ONLY ONE CROSS SPATIAL-SCALE BLOCK. "SHIFT WINDOW" MEANS THAT POSSESSES FOUR CROSS SPATIAL-SCALE BLOCKS WITH DIFFERENT SHIFT VALUES

spectral block	spatial block	shift window	PSNR(↑)	SSIM(↑)	SAM(↓)	ERGAS(↓)	SCC(↑)
✗	✗	✗	43.072	0.972	0.017	1.589	0.931
✓	✗	✗	46.515	0.984	0.013	1.097	0.960
✗	✓	✗	47.609	0.988	0.012	0.925	0.969
✗	✓	✓	48.706	0.991	0.010	0.837	0.974
✓	✓	✗	48.405	0.989	0.011	0.880	0.973
✓	✓	✓	**49.259**	**0.992**	**0.010**	**0.788**	**0.976**

TABLE II

AVERAGE QUANTITATIVE RESULT ON THE **ZY-M** DATASET WITH THE DIFFERENT BAND NUMBERS OF THE EXTRACTED "HIGH-LEVEL FEATURE"

band number	PSNR(↑)	SSIM(↑)	SAM(↓)	ERGAS(↓)	SCC(↑)
16	46.893	0.986	0.012	1.038	0.960
32	48.124	0.990	0.011	0.889	0.970
64	**49.259**	**0.992**	**0.010**	**0.788**	**0.976**

TABLE III

AVERAGE QUANTITATIVE RESULT ON THE **ZY-M** DATASET WITH THE DIFFERENT "RESNET BLOCKS" IN THE "HIGH-LEVEL FEATURE ENCODING BLOCK"

number of 'resnet' block	PSNR(↑)	SSIM(↑)	SAM(↓)	ERGAS(↓)	SCC(↑)
3	49.188	0.992	0.010	0.789	0.973
4	49.080	0.992	0.010	0.799	0.973
5	**49.259**	**0.992**	**0.010**	**0.788**	**0.976**
6	46.175	0.984	0.013	1.109	0.958

TABLE IV

AVERAGE QUANTITATIVE RESULT ON THE **ZY-M** DATASET WITH DIFFERENT UNFOLDING PATCH SIZES AND SHIFT VALUES IN THE CROSS SPATIAL-SCALE BLOCK

unfolding patch size	shift value	PSNR(↑)	SSIM(↑)	SAM(↓)	ERGAS(↓)	SCC(↑)
2	1	**49.259**	**0.992**	**0.010**	**0.788**	**0.976**
3	1	48.615	0.990	0.010	0.835	0.973
4	2	46.849	0.985	0.012	1.039	0.963
5	2	46.941	0.986	0.012	1.028	0.964
kernel size 2	stride 1	47.134	0.988	0.011	0.916	0.968

TABLE V

DIFFERENT LOSS FUNCTION COMBINATIONS AND THEIR QUANTITATIVE RESULTS ON THE **ZY-M** DATASET

L_1	L_{SSIM}	L_{ida}	PSNR(↑)	SSIM(↑)	SAM(↓)	ERGAS(↓)	SCC(↑)
✓	✗	✗	48.689	0.989	0.010	0.863	0.971
✓	✓	✗	48.393	0.991	0.010	0.861	0.972
✓	✓	✓	**49.259**	**0.992**	**0.010**	**0.788**	**0.976**

D. Ablation Study

In this section, we perform the ablation studies on the ZY-m dataset to demonstrate the effectiveness of each part of the network structures and loss functions.

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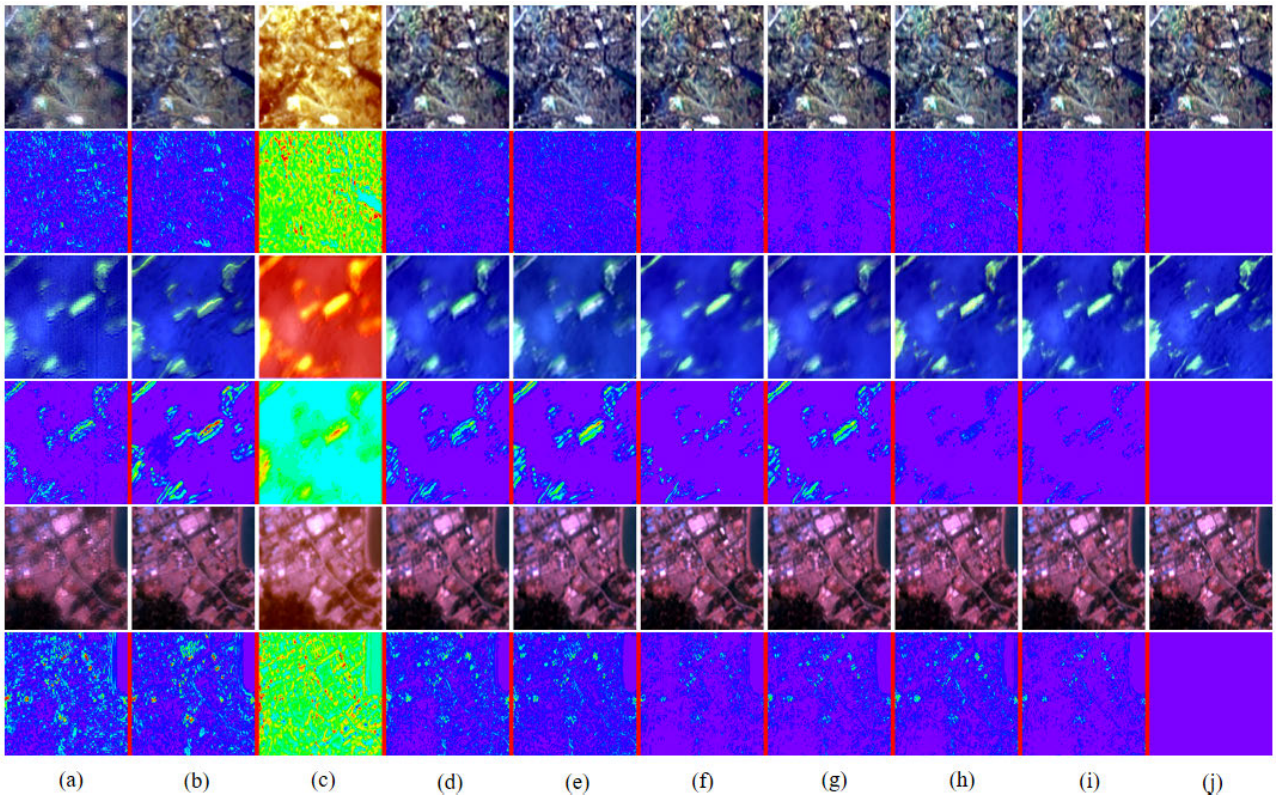


Fig. 8. Fused results (odd row) and error maps (even row) of different methods on ZY-m datasets. (a) SFIM. (b) CNMF. (c) SSRNet. (d) HSRNet. (e) EDBIN. (f) Fusformer. (g) MSSJFL. (h) SpfNet. (i) Ours. (j) Ground truth.

TABLE VI
DIFFERENT LOSS FUNCTION WEIGHTS [α AND β IN (13)] AND THEIR QUANTITATIVE RESULTS ON THE **ZY-M** DATASET

L_1	α	β	PSNR(↑)	SSIM(↑)	SAM(↓)	ERGAS(↓)	SCC(↑)
1	1	1	49.096	0.991	0.010	0.788	0.974
1	1	0.1	48.567	0.991	0.010	0.832	0.974
1	0.1	0.1	**49.259**	**0.992**	**0.010**	0.788	**0.976**
1	0.1	0.01	49.224	0.991	0.010	**0.787**	0.974
1	0.01	0.01	48.771	0.991	0.010	0.820	0.973

1) Network Structures: To demonstrate the effectiveness of the proposed cross spectral-scale block and shift-window-based cross spatial-scale block, we remove these modules and the experimental results are shown in Table I. The importance of the "cross spectral-scale" block can be inferred from the first two rows of this table. Note that the first row of Table I results from the network structure with only the "feature aggregation module."

Also, the effectiveness of the single cross spatial-scale block could be verified from the first and third rows. With the "shift-window" design (with four cross spatial-scale blocks), the quality indices all improve further, as shown in the third and fourth rows of Table I, which verifies its rationality and effectiveness. Overall, the proposed cross-scale blocks and the shift-window design all contribute to the final excellent fusion performance.

2) Shift-Window-Based Cross Spatial-Scale Block: In this section, to achieve the best fusion performance, we test the different hyperparameter settings in the proposed cross spatial-scale block.

First, we test the different numbers of the high-level feature bands and the "Resnet blocks" in the "high-level feature encoding block." The quantitative results on the ZY-m dataset are shown in Tables II and III. Theoretically, the more high-level feature bands, the more detailed and accurate the extracted semantic information would be. The experimental results verify this theory and band number 64 achieves the best results than 16 and 32. We also test the different numbers of "Resnet blocks" that represent the different network depths, to extract the semantic information as accurately as possible. The experimental results in Table III show that the five "Resnet blocks" achieve the best fusion performance.

In addition, we test the different unfolding patch sizes and shift values in the cross spatial-scale block to achieve the best balance between local feature learning and global similarity computation. As shown in Table IV, the unfolding patch size 2 and shift value 1 achieve the best fusion performance. Actually, this is similar to only one cross spatial-scale block (as in Fig. 3) with unfolding kernel size 2 and stride 1 [see (7)]. However, the results in the fifth row of Table IV show that with four spatial blocks, the stride equal to the unfolding size is better than the single block with stride 1. This is mainly caused by the transpose convolution operation in the cross spatial-scale block, which could use the multilayer nonlocal similarity (with different shift values) to reconstruct the HRHSI feature.

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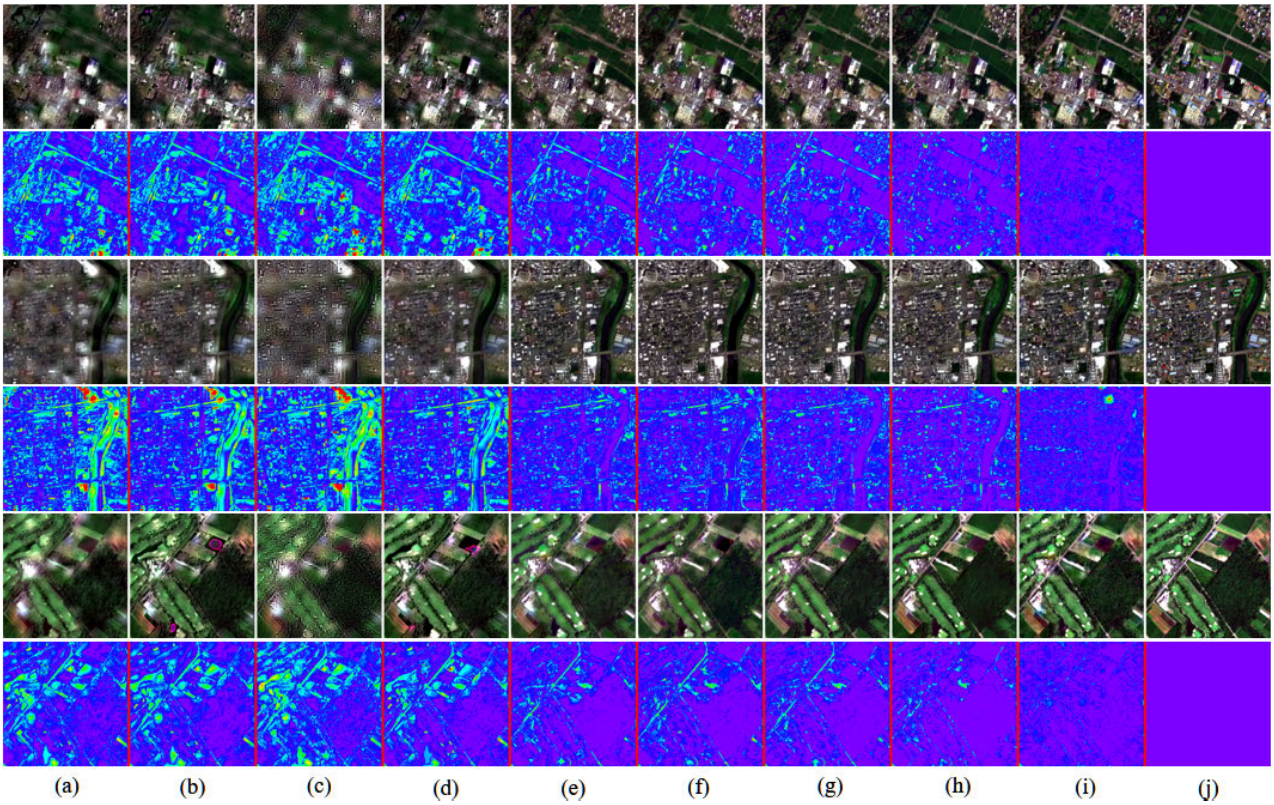


Fig. 9. Fused results (odd row) and error maps (even row) of different methods on Chikusei-p datasets. (a) GSA. (b) SFIM. (c) Wavelet. (d) MTF_GLP_HPM. (e) HSpeSet2. (f) GASN. (g) Pgnet. (h) RHDN. (i) Ours. (j) Ground truth.

3) Loss Functions: In this section, we test the significance of the proposed loss functions. First, we remove each loss function and the experimental results are listed in Table V. As shown in the last two rows of Table V, after the addition of L_{SSIM} and L_{ida}, the quality indices have all improved, especially the "PSNR" and "ERGAS" indices that improve by 1.17% and 8.69%, respectively, demonstrating the importance of these two loss functions. Note that the addition of L_{SSIM} slightly decreases the PSNR index, as shown in the second row of Table V, but the other indices have all improved, which could verify its effectiveness.

Furthermore, to achieve the best fusion performance, we test the different weighting coefficients of each loss function. The experimental results are shown in Table VI. Since the L_1 loss function plays the most important role, as shown in Table V, its weight is set to 1. As for the weights of L_{SSIM} and L_{ida}—α and β, the weights of 0.1 reach the first rank in four indices except for the ERGAS, as shown in the third row of Table VI. Thus, the best weighting coefficients for these loss functions are 1, 0.1, and 0.1.

E. Compared With SOTA Methods on HM-Fusion Task

In this section, we compare the proposed fusion network with other methods on the HM-fusion task using the Chikusei-m and ZY-m datasets. The visually fused results of all methods are shown in Figs. 7 and 8. Apparently, our method achieves the best spectral fidelity and spatial detail restoration on these two datasets. For the ZY-m dataset, the SSRNet method shows the spectral artifact, which can be seen from the fusion result and error map. Some methods suffer from the blurring effect such as SFIM, as shown in the first

TABLE VII
AVERAGE QUANTITATIVE RESULT ON THE CHIKUSEI-M DATASET

Method	PSNR(↑)	SSIM(↑)	SAM(↓)	ERGAS(↓)	SCC(↑)
SFIM	26.921	0.666	0.217	20.234	0.047
CNMF	42.958	0.979	0.049	4.075	0.973
SSRNet	42.838	0.977	0.037	2.547	0.977
HSRNet	47.070	0.992	0.034	2.591	0.985
EDBIN	45.235	0.987	0.043	3.254	0.978
Fusformer	47.585	0.987	0.035	4.215	0.979
MSSJFL	48.937	0.994	0.029	2.387	0.986
SpfNet	48.248	0.993	0.034	2.681	0.986
Ours	**51.052**	**0.996**	**0.023**	**1.956**	**0.991**

TABLE VIII
AVERAGE QUANTITATIVE RESULT ON THE ZY-M DATASET

Method	PSNR(↑)	SSIM(↑)	SAM(↓)	ERGAS(↓)	SCC(↑)
SFIM	42.050	0.940	0.022	1.855	0.924
CNMF	39.430	0.939	0.051	8.436	0.893
SSRNet	44.385	0.978	0.016	1.788	0.918
HSRNet	43.419	0.968	0.018	2.208	0.888
EDBIN	39.786	0.967	0.028	4.776	0.893
Fusformer	46.018	0.971	0.015	5.683	**0.982**
MSSJFL	45.386	0.981	0.014	1.300	0.943
SpfNet	46.010	0.981	0.015	1.279	0.946
Ours	**49.259**	**0.992**	**0.010**	**0.788**	0.976

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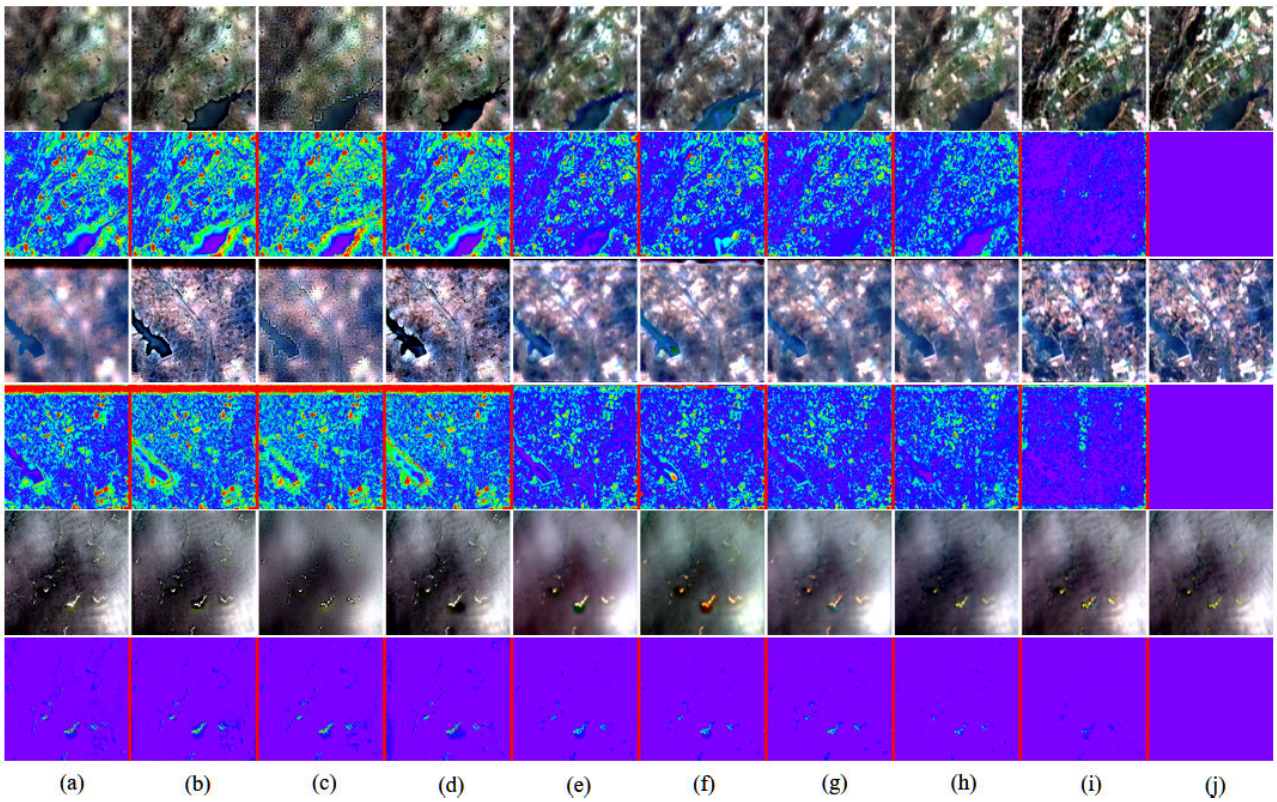


Fig. 10. Fused results (odd row) and error maps (even row) of different methods on ZY-p datasets. (a) GSA. (b) SFIM. (c) Wavelet. (d) MTF_GLP_HPM. (e) HSpeSet2. (f) GASN. (g) Pgnet. (h) RHDN. (i) Ours. (j) Ground truth.

TABLE IX
AVERAGE QUANTITATIVE RESULT ON THE CHIKUSEI-P DATASET

Method	PSNR(↑)	SSIM(↑)	SAM(↓)	ERGAS(↓)	SCC(↑)
GSA	31.149	0.817	0.141	2.921	0.649
SFIM	31.539	0.858	0.144	3.822	0.679
Wavelet	30.929	0.819	0.142	3.237	0.614
MTF_GLP_HPM	31.958	0.865	0.146	3.835	0.726
HSpeSet2	36.113	0.924	0.103	1.806	0.887
GASN	35.382	0.909	0.124	1.997	0.881
Pgnet	36.486	0.932	0.095	1.684	0.898
RHDN	38.016	0.946	0.079	1.490	0.924
Ours	**39.034**	**0.956**	**0.078**	**1.413**	**0.935**

TABLE X
AVERAGE QUANTITATIVE RESULT ON THE ZY-P DATASET

Method	PSNR(↑)	SSIM(↑)	SAM(↓)	ERGAS(↓)	SCC(↑)
GSA	33.526	0.847	0.051	1.404	0.689
SFIM	33.901	0.847	0.047	1.394	0.706
Wavelet	33.909	0.803	0.050	2.649	0.605
MTF_GLP_HPM	34.293	0.853	0.045	1.366	0.741
HSpeSet2	38.464	0.915	0.037	2.287	0.868
GASN	37.212	0.920	0.042	0.782	0.859
Pgnet	38.434	0.927	0.037	0.724	0.894
RHDN	39.482	0.934	0.032	0.577	0.922
Ours	**43.205**	**0.974**	**0.019**	**0.380**	**0.952**

TABLE XI
AVERAGE QUANTITATIVE RESULT ON THE XIONGAN DATASET

Method	PSNR(↑)	SSIM(↑)	SAM(↓)	ERGAS(↓)	SCC(↑)
GSA	32.326	0.913	0.062	0.742	0.721
SFIM	33.591	0.946	0.057	0.657	0.793
Wavelet	31.749	0.870	0.069	0.840	0.665
MTF_GLP_HPM	34.841	0.951	0.055	0.568	0.847
HSpeSet2	32.469	0.944	0.103	0.840	0.888
GASN	33.856	0.940	0.082	0.695	0.804
Pgnet	35.424	0.960	0.053	0.524	0.863
RHDN	38.026	0.969	0.052	0.386	0.924
Ours	**39.921**	**0.977**	**0.035**	**0.311**	**0.938**

column of Figs. 7 and 8. In contrast, our method shows the best spatial detail restoration, as shown in the fusion results and error maps (such as the white building in the first row of Fig. 7 and the island in the third row of Fig. 8).

The quantitative results in Tables VII and VIII also show that our method achieves absolute superiority over other methods on five quality indices. For the Chikusei-m dataset, the SFIM method obtains the worst result, which is consistent with the visual results in Fig. 7. Also, our method achieves the first rank in all five indices, especially the PSNR, SAM, and ERGAS indices. For the ZY-m dataset, the PSNR and SSIM indices of our results are better than those of the suboptimal results by 6.97% and 1.12%, respectively. Although the SCC index of our method is inferior to the Fusformer method, this index difference is marginal and the other indices of this method are worse than ours.

F. Compared With SOTA Methods on HP-Fusion Task

We also compare the proposed fusion method with other methods on the HP-fusion task using the Chikusei-p, ZY-p,

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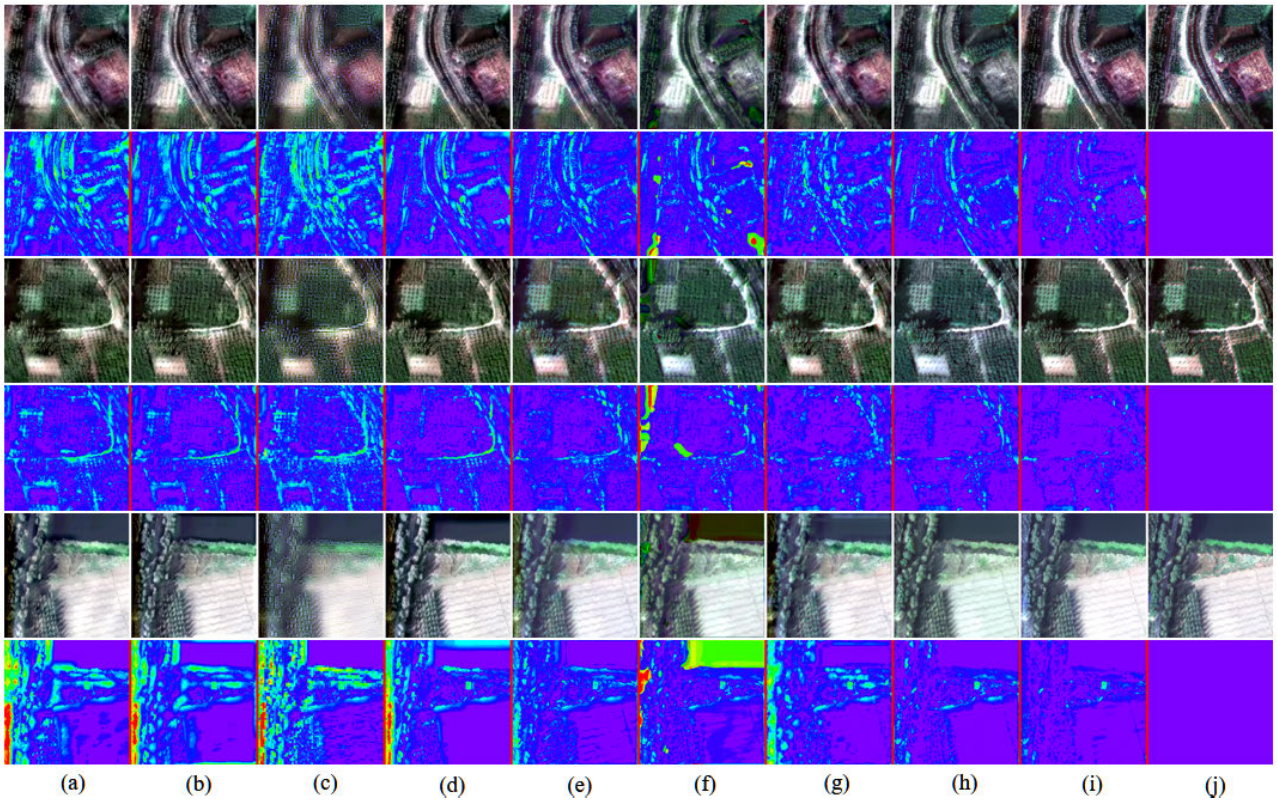


Fig. 11. Fused results (odd row) and error maps (even row) of different methods on XiongAn datasets. (a) GSA. (b) SFIM. (c) Wavelet. (d) MTF_GLP_HPM. (e) HSpeSet2. (f) GASN. (g) Pgnet. (h) RHDN. (i) Ours. (j) Ground truth.

TABLE XII
AVERAGE INFERENCE TIME AND MODEL PARAMETERS ON THE CHIKUSEI-P DATASET

Method	Time(s)(↓)	Parameters(m)(↓)
GSA	0.882	—
SFIM	7.021	—
Wavelet	0.563	—
MTF_GLP_HPM	2.250	—
HSpeSet2	0.004	0.118
GASN	0.022	2.003
Pgnet	0.026	0.049
RHDN	0.123	4.803
Ours	0.039	2.915

and XiongAn datasets. The qualitative fusion results are shown in Figs. 9–11. Similar to the results in the HM-fusion task, our method still achieves the best fusion performance in terms of spectral and spatial fidelity. Due to the high ill-posedness of the HP-fusion task with the huge fusion ratio, the traditional methods all suffer from the blurring effect, especially at the edges of buildings and roads. This can also be seen from the error maps in the even rows of Figs. 9 and 10. Even for the XiongAn dataset with the smoother texture, the traditional methods still suffer from the blurring effect, such as the wavelet method's fused results, as shown in the third column of Fig. 11.

The DL-based methods generally perform better than the traditional methods in terms of spectral and spatial fidelity. However, most of them have not considered the huge spatial and spectral resolution difference between the input images, and their fusion performance is hard to improve further. Also, our proposed method achieves the best spectral fidelity and spatial detail restoration than other methods.

The quantitative results also confirm the superiority of the proposed method in restoring the spectral and spatial details of HRHSI. As shown in Tables IX–XI, our method achieves the first rank in all five indices. For example, our PSNR and SSIM indices are better than the second rank by 2.68% and 1.06%, 9.43% and 4.28%, and 4.98% and 0.83% on these three datasets.

Overall, the fusion results on both HM-fusion and HP-fusion tasks demonstrate the superiority and robustness of the proposed fusion network.

G. Comparison of Computational Complexity

In this section, we list the inference time and the number of parameters of all HP-fusion methods to compare their fusion efficiency. This computation is performed on the Chikusei-p dataset.

As shown in Table XII, the traditional methods all suffer from large time consumption, especially the SFIM and MTF_GLP_HPM methods. In comparison, the DL-based methods all have an immediate inference time. Despite the fast inference time and fewer parameters of the HSpeSet2, GASN,

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and Pgnet methods, their fusion performance is inferior to ours according to the above experimental results. Also, the increase in inference time of our method compared to other DL-based methods is negligible. Therefore, it can be concluded that our method achieves the best balance between the fusion performance and the model complexity.

V. Conclusion

In this article, to adapt to the huge spectral and spatial resolution difference between LRHSI and HRMPI, we design the cross spectral-scale and SW-CSNA blocks. These two blocks could capture the global and patch-level similarity to combine the advantages of nonlocal and local feature learning. In addition, to enforce the correctness of the extracted high-level semantic feature, we design an ida loss function with fast computation speed. Experimental results on simulated and real datasets demonstrate the effectiveness and superiority of the proposed fusion network.

Although the proposed fusion network achieves excellent fusion performance, the model trained on one dataset is difficult to be used on another dataset. This is caused by the inconsistent number of bands and different image features. Also, this is our research work in the future.

Acknowledgment

The authors would like to thank Baidu AI Studio for providing the computing power support.

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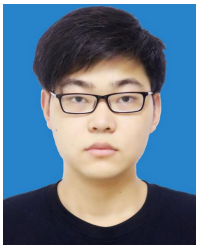
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